

Structural and Transport Properties of Li/S Battery Electrolytes: Role of the Polysulfide Species

Chanbum Park,^{†,§} Arne Ronneburg,^{‡,§} Sebastian Risse,[‡] Matthias Ballauff,^{‡,§} Matej Kanduč,^{†,||} and Joachim Dzubiella^{*,†,⊥}

[†]Research Group for Simulations of Energy Materials and [‡]Institut für Weiche Materie und Funktionale Materialien, Helmholtz-Zentrum Berlin für Materialien und Energie, Hahn-Meitner-Platz 1, D-14109 Berlin, Germany

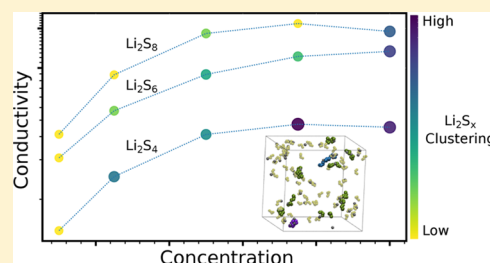
[§]Institut für Physik, Humboldt-Universität zu Berlin, Newtonstr. 15, D-12489 Berlin, Germany

^{||}Jožef Stefan Institute, Jamova 39, SI-1000 Ljubljana, Slovenia

[⊥]Applied Theoretical Physics—Computational Physics, Physikalisches Institut, Albert-Ludwigs-Universität Freiburg, Hermann-Herder Strasse 3, D-79104 Freiburg, Germany

S Supporting Information

ABSTRACT: Lithium–sulfur (Li/S) batteries are regarded as one of the most promising energy-storage devices beyond lithium-ion batteries because of their high energy density of 2600 W h kg^{−1} and the affordable cost of sulfur. Meanwhile, some challenges inherent to Li/S batteries remain to be tackled, for instance, the polysulfide (PS) shuttle effect, the irreversible solidification of Li₂S, and the volume expansion of the cathode material during discharge. On the molecular level, these issues originate from the structural and solubility behavior of the PS species in bulk and in the electrode confinement. In this study, we use classical molecular dynamics simulations to develop a working model for PS of different chain lengths in applied electrolyte solutions of lithium bistriflimide (LiTFSI) in 1,2-dimethoxyethane and 1,3-dioxolane mixtures. We investigate conductivities, diffusion coefficients, solvation structures, and clustering behavior and verify our simulation model with experimental measurements available in the literature and newly performed by us. Our results show that diffusion coefficients and conductivities are significantly influenced by the chain length of PS. The conductivity contribution of short chains like S₄^{2−} is lower than that of longer PS chains such as S₆^{2−} or S₈^{2−} despite the fact that the diffusion coefficient of S₄^{2−} is higher than that for longer PS chains. The low conductivity of Li₂S₄ can be attributed to its low degree of dissociation and even to the formation of large clusters in the solution. It is also found that an addition of 1 M LiTFSI into PS solutions considerably reduces the clustering behavior. Our simulation model enables future systematic studies in various solvating and confining systems for the rational design of Li/S electrolytes.



1. INTRODUCTION

Lithium-ion batteries are the dominant type of rechargeable batteries currently on the market. Relatively efficient battery life cycle as well as decent energy and power densities make lithium-ion batteries attractive. Meanwhile, the growing demand for large-scale energy-storage systems and electric vehicles motivates the battery community to seek new solutions with higher energy density and lower material costs. One of the candidates is lithium–sulfur (Li/S) batteries owing to the low cost of sulfur, high energy density of 2600 W h kg^{−1}, and high theoretical specific capacity of 1675 A h kg^{−1}.^{1,2}

In spite of these attractive advantages, a practical Li/S battery performance has not been realized. Unlike lithium-ion batteries, where intercalation of Li⁺ provides the major dynamics, elemental sulfur S₈ in the cathode chemically reacts with Li⁺.^{3,4} During discharge, Li⁺ migrates from the lithium anode through the electrolyte and the separator to the sulfurized cathode. Subsequently, solid sulfur converts into

linear chainlike polysulfide (PS) intermediates, whose length disproportionately shrinks upon reduction with lithium until Li₂S finally precipitates.^{2,4–6} During charging, the reverse chemical reaction partially takes place. As a result of a complex cell chemistry, Li/S cells have to face two major challenges. First, parasitic side reactions at the lithium electrode can occur that originate from irreversible reduction of PS to lithium sulfide.^{4,6,7} This shuttle phenomenon, originating from the migration of PS from the cathode materials toward the anode, is one source of capacity fading. Second, during discharge, Li₂S₂ increases in volume by 80%, which can lead to mechanical stress in the cathode material and loss of contact to the conductive host matrix.^{1,8,9} These adverse phenomena are the major bottlenecks of the Li/S battery development. As indicated by recent work,^{2,10} extensive efforts have been

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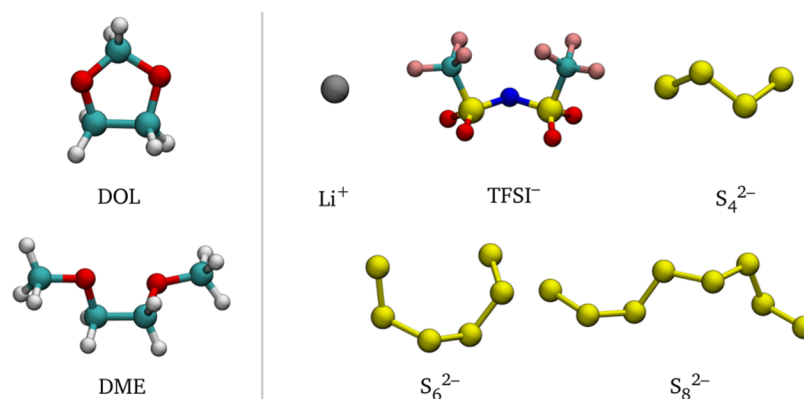


Figure 1. Solvent molecules 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME), and ions Li^+ , bis(trifluoromethanesulfonyl)imide (TFSI^-), and polysulfides (PS) with chain lengths $x = 4, 6$, and 8 considered in this study.

devoted into minimizing the shuttling of the PS by confining them into meso- or microporous materials.^{11–15} Because the soluble intermediate PS species cause the shuttling, the reaction mechanism pathway can also be altered by choices of the solvent and ions and thus changing the stability of the intermediate species, charge/discharge rate, etc.^{4,15–20} Therefore, more fundamental and systematic studies on the structural and dissolution properties of PS in various applied solvents are in urgent need.

The most frequently formed intermediates during the battery cycling process are S_x^{2-} (with $x = 4, \dots, 8$). These PS ionic chains in the electrolyte can merge with Li^+ into clusters such as $(\text{Li}_2\text{S}_x)_n$ (with $n \geq 1$).²¹ The clusters of Li_2S_4 are believed to be the last polysulfide intermediates before the formation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ insoluble aggregates during the discharge process.²² In recent years, various quantum chemistry calculations and classical molecular dynamics (MD) models have demonstrated poor solubility and thus the tendency for cluster formation of shorter PS species.^{21–29}

The solubility of PS and the size distribution of the PS clusters and their morphologies depend on the properties of the solvent and added electrolyte,^{30–33} current density, and temperature.³⁴ A recent study by Andersen et al.³⁵ investigated PS clustering using classical MD simulations, density functional theory calculations, and experiments. Another classical MD simulation by Rajput et al.²⁸ suggested that introducing TFSI^- ions, which compete with PS for Li^+ , weakens the PS-Li^+ clustering networks, resulting in higher solubility of PS. However, the diffusion coefficients in that study differ from those measured by the pulse-field gradient nuclear magnetic resonance (PFG-NMR) methods by more than 1 order of magnitude.²⁸ The control of clustering may thus represent the critical objective for battery performance. It is thus of utmost importance to obtain deeper understanding of the structural and transport properties of these highly complex, multi-component PS solutions.

Our endeavor of this and a preceding study³⁶ is to develop an efficient yet accurate force field for purely classical molecular dynamics simulations of Li/S battery electrolytes. We have successfully described the static and transport properties of the solutions of 1,2-dimethoxyethane (DME)/1,3-dioxolane (DOL) containing LiTFSI and LiNO_3 salts.³⁶ In this study, we implement PS components into the previously developed model.³⁶ For the most relevant PS chain lengths of four, six, and eight sulfur atoms, we investigate the clustering

behavior of PSs, as well as their solvation structure, diffusion, and conductivity properties.

2. METHODS

2.1. Model and Force Fields. The solvent in our atomistic model is a 1:1 molar mixture of 1,2-dimethoxyethane (DME) and 1,3-dioxolane (DOL), which contains different amounts of Li^+ and bis(trifluoromethanesulfonyl)imide ions (TFSI^-), as well as three different kinds of PS ions: S_4^{2-} , S_6^{2-} , and S_8^{2-} (see Figure 1).

We employ the OPLS-AA force field^{37,38} for DME and DOL, the CL&P force field for TFSI^- ,³⁹ and Dang et al.⁴⁰ for the Li^+ ion, together with the geometric combination rules for Lennard-Jones (LJ) interactions. The Coulomb interactions of ions are treated by the electronic continuum correction (ECC) method,^{41–47} which takes the electronic polarizability into account implicitly. In this approach, formal ionic charges q_i in the interaction Hamiltonian are replaced by effective, rescaled values q_i^{eff}

$$q_i^{\text{eff}} = \frac{q_i}{\sqrt{\epsilon_\infty}} \quad (1)$$

where $\epsilon_\infty = 1.93$ is the high-frequency dielectric permittivity of the bare solvent.³⁶ This approach turned out to be very successful for many structural and dynamic properties of $\text{LiTFSI}/\text{LiNO}_3$ salts in DME/DOL solutions³⁶ and is applied in this study also to the PS ions.

We devote special attention to the parametrization of the PS ions. The bonded parameters for PS ions are taken from the recent work by Rajput et al.²⁸ To obtain the partial charges, we performed quantum mechanical calculations using the electrostatic potential surface method implemented in the GAUSSIAN09 package⁴⁸ with the B3LYP functional at the aug-cc-pvdz basis set level. The resulting (unscaled) and the corresponding effective (rescaled via eq 1) partial charges of sulfur atoms in PS are summarized in Table 1.

Different LJ parameters for S atoms have been tested using various force fields: OPLS-AA,³⁷ AMBER99,⁴⁹ CHARMM,⁵⁰ ENCAD,⁵¹ ECEPP,⁵² UFF,⁵³ and DREIDING⁵⁴ (see Table S1 in the Supporting Information (SI)). We finally opted for the DREIDING force field for the S atoms (parameterized to reproduce S_8 crystal properties). However, as it turns out, the LJ interaction size $\sigma_{\text{S-Li}^+}$ between a terminal S atom and Li^+ is a critical parameter that determines the degree of aggregation and clustering propensities, as well as conductivity of PS ions.

Table 1. Formal (q_i) and Effective (q_i^{eff} , Computed via Equation 1) Partial Charges of Terminal and Internal S Atoms in the PS Chains^a

	terminal S		internal S	
	q_i/e	q_i^{eff}/e	q_i/e	q_i^{eff}/e
S_4^{2-}	-0.7702	-0.5546	-0.2298	-0.1655
S_6^{2-}	-0.6537	-0.4707	-0.1731	-0.1247
S_8^{2-}	-0.6223	-0.4481	-0.1259	-0.0907

^a e is the elementary charge.

Therefore, we tuned the LJ parameter manually to $\sigma_{S-Li^+} = 0.275$ nm to reproduce experimental conductivity values. The procedure and the details are provided in SI (Figures S1 and S2).

2.2. Simulation Details and Protocols. All-atom MD simulations are carried out with the GROMACS 5.1 simulation package.⁵⁵ Initial simulation structures of ions in the solvent are constructed with the PACKMOL package.⁵⁶ The molecules are randomly inserted into the simulation box and undergo an energy minimization, followed by a simulated annealing from 440 K down to 298 K within a time interval of 3 ns in the NVT ensemble. The production simulations are carried out in the NpT ensemble at a constant pressure of 1 bar and temperature of 298 K. The pressure is controlled by the Parrinello–Rahman barostat with a time constant of 2 ps, whereas the temperature is controlled by the Berendsen thermostat with a time constant of 0.1 ps.⁵⁷ The LJ potentials are truncated at 1.3 nm. The particle mesh Ewald method with a Fourier spacing of 0.12 nm and a 1 nm real-space cutoff are used for calculating electrostatic interactions. The LINCS algorithm is used for all bond constraints.

2.3. Analysis. **2.3.1. Dielectric Constant.** The non-electronic part of the static dielectric constant in the simulations, ϵ_{MD} , is evaluated from the fluctuations of the system's dipole moment (excluding ionic monopoles) $\mathbf{M}_{\text{MD}}$ (using rescaled charges)^{58,59}

$$\epsilon_{\text{MD}} = 1 + \frac{1}{3V\epsilon_0 k_B T} (\langle \mathbf{M}_{\text{MD}}^2 \rangle - \langle \mathbf{M}_{\text{MD}} \rangle^2) \quad (2)$$

where V , k_B , and T stand for the volume of the simulation box, the Boltzmann constant, and the absolute temperature, respectively. Since we are employing the ECC approach, the total effective dielectric constant is

$$\epsilon = \epsilon_{\infty} \epsilon_{\text{MD}} \quad (3)$$

More details can be found in ref 36.

2.3.2. Diffusion Coefficients. The self-diffusion coefficients of ions and molecules are computed via the mean square displacement relation

$$D_{\text{MD}} = \lim_{\Delta t \rightarrow \infty} \frac{\langle r^2(\Delta t) \rangle}{6\Delta t} \quad (4)$$

In addition, the finite-size effects of the simulation box are corrected by extrapolating measured D_{MD} values at different box sizes L to $L \rightarrow \infty$ (see Figure S3 in SI).

2.4. Conductivity. Conductivities of the solutions in our simulations are computed from Ohm's law

$$J = \kappa E \quad (5)$$

where E is the applied electric field and $J = \sum J_i = \sum \kappa_i E$ is the resulting total current density. This relation defines partial

ionic conductivities κ_i for each individual ionic species. The conductivities κ_i are evaluated from the linear slope of J versus E in the linear-response regime (see Figure S1 in SI).

2.4.1. Clustering. We define a cluster of PS ions as a group of those PS ions whose at least one of the terminal S atoms is separated from a terminal S atom of any other PS ion in the cluster by less than $r_0 = 0.53$ nm. The cutoff value r_0 is chosen as the first minimum after the main peak of the radial distribution function (RDF) of terminal S atoms and thus corresponds to the distance between two terminal S atoms that have a bridging Li^+ ion in between (see Figure S4 in SI). By ensemble averaging of clusters in the simulations, we obtain an equilibrium cluster size distribution $P(N)$, where N is the number of PS ions in a cluster. The statistical uncertainties are calculated with the block averaging procedure.

2.4.2. Coordination Number and Dissociated Li^+ Ion. The coordination number N_{ij} of the molecule of type i that is surrounded by molecules of type j is computed from the RDF as⁶⁰

$$N_{i,j} = 4\pi c_j \int_0^{R_M} g_{ij}(r) r^2 dr \quad (6)$$

where R_M is the distance of the first minimum after the first peak in the RDF and c_j is the bulk concentration of molecules of type j .

2.5. Experimental Methods. Electrolyte mixtures were prepared by adding the corresponding stoichiometric ratio of lithium sulfide (Li_2S) and elemental sulfur (S_8) to a mixture of DME and DOL (1:1 by mole). The uncertainty in the composition of the electrolyte is less than 5%. Although the various disproportion reactions of lithium polysulfides in solution are known, it is assumed that the majority of PS appears in the length as prepared by the intended stoichiometric ratio, as shown by Barchasz et al., using high-performance liquid chromatography.⁶¹ The gravimetric density of the electrolyte mixtures was measured at 18 °C and 1 bar using a chempro/PAAR DMA 602 density meter. The average value of 10 measurements of the natural frequency of a glass tube filled with the solution was taken to calculate the density. Millipore water and air served as reference for this calculation. The error on the values is 2×10^{-5} . The viscosity was determined using a capillary viscometer (SI Analytics S0101/0a) and a laboratory stopwatch. The measurements were performed in an argon-filled glovebox at 25 °C. The error of the measurement varies from 0.07 to 1%. The conductivity was evaluated by performing impedance spectroscopy in the frequency range of 100 mHz to 1 MHz with 5 mV root mean square voltage signal and 20 points per decade. Three different direct current voltages (0, 1, and 1.5 V) were used. A GAMRY interface 1000 potentiostat was used. The cell was a standard CR 2032 coin cell. Two stainless steel plates were used as electrodes. A PTFE ring (thickness, 0.27 mm; diameter, 10 mm) was used to adjust the distance between the two electrodes. The conductivity was determined by the intersection of the impedance with the x -axis at high frequencies in a Nyquist plot.

3. RESULTS

3.1. Density and Dielectric Constant. We start our analysis by calculating the density of various PS in DME/DOL solutions, the results of which are plotted in Figure 2a. Universally, the density is an increasing function of the Li_2S_x concentration for all PS types. This rise, as well as the

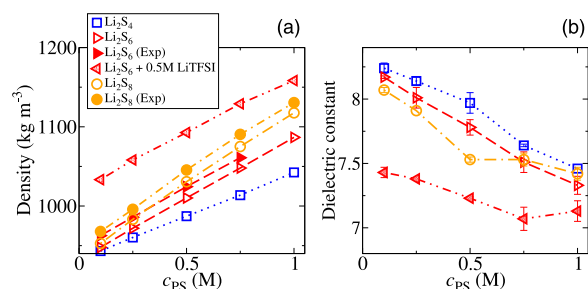


Figure 2. (a) Density and (b) dielectric constant as a function of polysulfide concentration in DME/DOL (1:1) solvent in MD simulations. Experimental measurements for Li_2S_6 and Li_2S_8 are depicted by solid triangles and circles in (a).

magnitude is in good accordance with our experimental measurements for Li_2S_6 and Li_2S_8 (solid triangles and circles, respectively; Figure 2a). Note that in our simulations, we only consider one-component (i.e., polydisperse) PS solutions. In reality, however, the monodispersity cannot be reached due to the disproportionation reactions of PS, but the majority of the PS should appear in the length as prepared.⁶¹

The dielectric constant as a function of Li_2S_x concentration is shown in Figure 2b. It decreases with ion concentration, as is also the case in other electrolytes.^{62–65} This decrement of the dielectric constant is caused by a local dielectric saturation. Namely, solvent molecules tend to strongly orient and anchor around an ion and do not contribute to the dielectric constant. As more ions are present in the solution, larger fraction of solvent molecules are part of the solvation shells of the ions and thus larger is the local dielectric saturation. The dielectric decrement is slightly larger for longer PS chains. The addition of 0.5 M LiTFSI into the Li_2S_6 solution decreases the dielectric constant considerably, which can be also explained by the local dielectric saturation due to added ions into the solution.

3.2. Solvation Structure and Radial Distribution Functions (RDFs). In the following, we take a look at the solvation structure of Li^+ ions. Figure 3 shows RDFs of various molecules and Li^+ ions in DME/DOL (a–c) and in DME/DOL with 1 M LiTFSI (d–f). The top two panels (a and d) show the distribution of terminal S atoms (S_{ter}) of PSs. The very high first peak in all of the cases indicates a strong binding affinity between Li^+ and PS ions and can be attributed to the electrostatic attraction between Li^+ and S_x^{2-} . Moreover, the height of the peak and thus the binding strength are diminishing with the length of the PS ions (when going from S_4^{2-} to S_8^{2-}). These trends are consistent with a recent classical MD simulation study.²⁸ This can be explained by stronger charge localization (of the net valency -2) at terminal ends of shorter PS ions.²⁶ As seen in Table 1, shorter chains have higher partial charges at the termini, thus facilitating the attraction with Li^+ . Furthermore, the geometry of the PS chains also plays a role in the solvation shell. Snapshots in Figure 4a,c,e show that short PS chains (e.g., S_4^{2-}) are able to tightly wrap around a Li^+ ion. Conversely, longer PS chains (e.g., Li_2S_6 and Li_2S_8) do not pack so tightly around the Li^+ .

The different binding strengths result in different compositions of the Li^+ solvation shell, which can be described by coordination numbers in Figure 5. Namely, shorter PS chains drive out other molecular species from the first hydration shell of Li^+ (a–d). Due to this PS packing in the first solvation shell of Li^+ , fewer DME or DOL molecules can populate the surrounding of Li^+ in the presence of Li_2S_4 than in the cases of

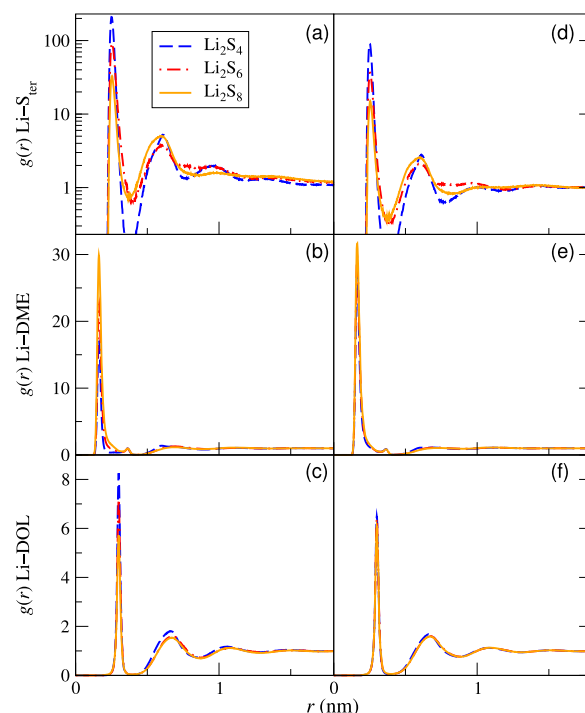


Figure 3. Radial distribution functions (at 0.25 M Li_2S_x) between Li^+ and (a, d) terminal S (S_{ter}) in S_x^{2-} ($x = 4, 6$, and 8); (b, e) DME; and (c, f) DOL in DME/DOL (a–c) and in DME/DOL with 1 M LiTFSI (d–f).

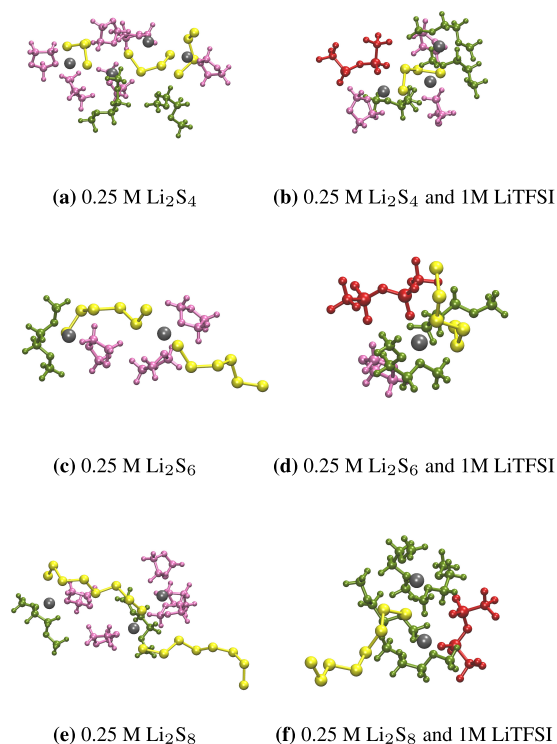


Figure 4. Snapshots of Li^+ solvation shell in DME/DOL and different amounts of ions. Color code: Li^+ (gray), S_x^{2-} (yellow), TFSI⁻ (red), DME (green), and DOL (pink).

Li_2S_6 or Li_2S_8 (lower peak in $g_{Li^+-DME}(r)$ in the case of Li_2S_4 than in Li_2S_8 in Figure 3b). The DME coordination number of Li^+ in Figure 5 increases from Li_2S_4 to Li_2S_8 . Ab initio MD simulations by Kamphaus²⁶ also showed similar trends. This

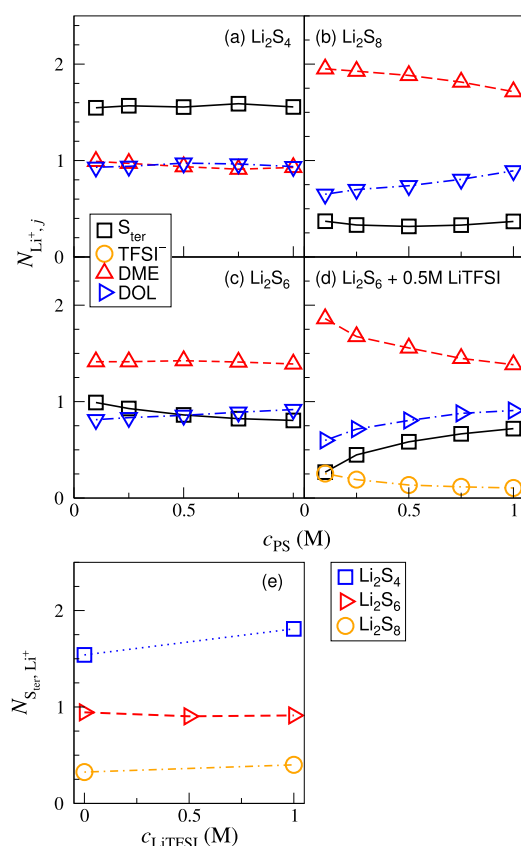


Figure 5. (a–d) Coordination numbers of molecules j around the Li^+ ion ($N_{Li^+,j}$) as a function of polysulfide concentration (see legend). (e) Li^+ coordination number around S_{ter} (N_{S_{ter}, Li^+}) as a function of LiTFSI concentration.

densely packed solvation structure by S_4^{2-} gives less chance for Li^+ in the solvation shell to contact with solvent molecules. It restricts the Li^+ exchange between S_4^{2-} and solvent molecules, resulting in lower solubility.^{26,66}

Now we investigate the effects of LiTFSI in the solution. Figure 3d shows that the magnitudes of the main peaks in $g_{Li^+-S_4^{2-}}(r)$ decrease after 1 M of LiTFSI is added (cf. panel a), as also consistent with previous studies²⁸ (also see $g_{Li^+-TFSI}(r)$ in Figure S5 in SI). An important insight can be gained from the Li^+ coordination number around a PS molecule shown in Figure 5e. Evidently, the number of Li^+ ions around S termini does not change upon introducing LiTFSI into the system. This means that LiTFSI does neither weaken the Li^+ –PS bonds nor do additional Li^+ ions from LiTFSI bind to PS. Thus, the decrease in the RDF peaks (Figure 3d) and the different solvation shell composition (Figure 5d) arrive on the expense of the added Li^+ ions.

3.3. Conductivity. In Figure 6, we show the ionic conductivity from the MD simulations and experimental measurements (ours, by Safari et al.,⁶⁷ and by Fan et al.²⁰) for PS concentrations in the range of 0.1–1 M. As seen, the experimental trends of the four studied systems (Li_2S_4 , Li_2S_6 , Li_2S_8 , and $Li_2S_6 + 0.5 M LiTFSI$) are well captured by the simulations. In the cases of Li_2S_6 and Li_2S_8 , the conductivities are by a factor of 3 higher than those in experiments. Nevertheless, we regard the results satisfactory as these quantities are extremely sensitive to the molecular interactions and thus prone to large errors, sometimes of more than an

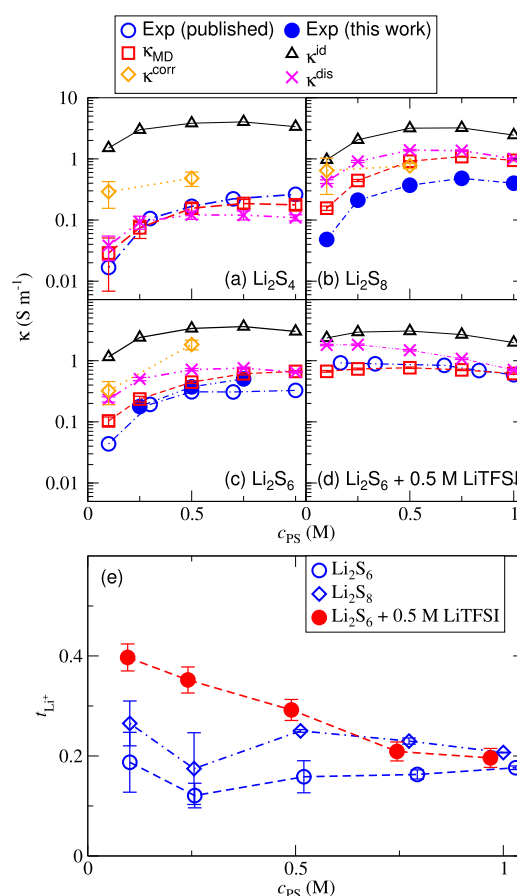


Figure 6. (a–d) Conductivities as a function of polysulfide concentration from MD simulations (squares), experiments (circles), and theoretical approaches [ideal ionic conductivity κ^{id} assuming eq 7 (triangles up), κ^{corr} using eq 8 (diamonds), and κ^{dis} using eq 9 (crosses)]. (a) Li_2S_4 : exp. by Safari et al.⁶⁷ (DME/DOL), (b) Li_2S_8 : exp. by us (DME/DOL), and (c) Li_2S_6 : exp. by Safari et al.⁶⁷ (empty circles) and by us (filled circles) (DME/DOL). (d) $Li_2S_6 + 0.5 M LiTFSI$: exp. by Fan et al.²⁰ (0.5 M LiTFSI DME/DOL). (e) Estimated Li^+ transference number from MD for different Li_2S_x concentrations (with/without LiTFSI).

order of magnitude. The conductivities from the simulations for the ternary electrolytes of $Li_2S_6 + LiTFSI$ are also congruent with experiments, capturing even the saturation and the decrease in the conductivity with increasing PS concentration above 0.5 M.

Ideal ionic conductivity in the limit of low concentrations is given by the Nernst–Einstein (NE) equation

$$\kappa^{id} = \frac{e^2}{k_B T} \sum_i z_i^2 c_i D_i \quad (7)$$

where z_i stands for the ion valency and c_i stands for the ion concentration of species i . Using the diffusivities D_i obtained from the simulations, we calculate the ideal conductivities in Figure 6 (triangles). Clearly, the values are an order of magnitude higher, which we attribute to substantial ionic pairing.^{20,36} Namely, eq 7 is a limiting law and thus neglects ion–ion correlations.

In a first-order correction to the ideal conductivity, the correlations can be perturbatively taken into account, which leads to⁶⁸

$$\kappa^{\text{corr}} = \kappa^{\text{id}} + \frac{2e^2}{3\eta} \sum_{i,j} z_i z_j c_i c_j \int_0^\infty h_{ij}(r) r \, dr \quad (8)$$

where η is the solvent viscosity. Here, the ion–ion correlation effects are expressed via $h_{ij}(r) = g_{ij}(r) - 1$. Equation 8 thus constitutes a useful structure–transport relationship, applicable at least for not too dense solutions. Taking $h_{ij}(r)$ from our simulations, we plot the correlation-corrected values κ^{corr} as diamond symbols in Figure 6. Evidently, the negative effect of ionic pairing is qualitatively captured by the correlation term and the values approach closer the MD results. Among all of the contributing ion pairs to the second term in eq 8, the most of contribution comes from the $\text{Li}^+ - \text{S}_x^{2-}$ pair (see $\int_0^\infty h_{ij}(r) r \, dr$ values in Table S2). This means that the strong binding between the latter two ions is the main reason for the observed low conductivity. Still, the correlation correction given by eq 8 cannot provide a fair quantitative agreement with the measured values, and hence we conclude that even at lower concentrations already, correlations beyond the pair level are important, e.g., from clustering effects (see further below).

Thus, in the limit of very strong ion pairing, where most of the ions are associated into neutral ion pairs and clusters, we can expect that only the dissociated ions contribute to the conductivity. In this simplified picture, we replace ionic concentrations c_i in the NE eq 7 by the concentrations of dissociated ions, $\omega_i c_i$

$$\kappa^{\text{dis}} = \frac{e^2}{k_B T} \sum_i z_i^2 \omega_i c_i D_i \quad (9)$$

where ω_i stands for the fraction of dissociated ions of type i . For simplicity of our treatment, we assume that only Li^+ and PS ions are involved in pairing, whereas TFSI^- ions are completely dissociated, $\omega_{\text{TFSI}^-} = 1$, as they only weakly bind with Li^+ . Li^+ and PS species are subject to the following equilibrium



We assume that further dissociation into $2\text{Li}^+ + \text{S}_x^{2-}$ is much less probable and we neglect it.

Figure 7 shows the fraction of dissociated Li^+ in various solutions, defined as the population of those Li^+ ions that do not have PS in their hydration shells. The dissociation is by far the lowest for Li_2S_4 and higher for Li_2S_6 and Li_2S_8 ; similar trends are also observed in a recent study.³⁵ Adding LiTFSI

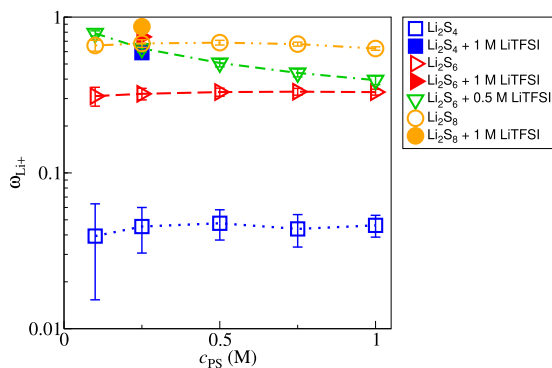


Figure 7. Fraction of dissociated Li^+ ions as a function of polysulfide concentration in different solutions. It is defined as the fraction of the Li^+ population without PS ions in their hydration shells.

increases the dissociated degree notably, which implies that the additional lithium from LiTFSI remains more or less dissociated. In cases without LiTFSI, the effective concentrations of Li^+ and $(\text{LiS}_x)^-$ species are both equal to $2c_{\text{PS}}\omega_{\text{Li}^+}$. Introducing LiTFSI adds equivalent concentrations of c_{TFSI^-} to Li^+ as well as TFSI^- species, since the added Li^+ are all dissociated. Thus, the modified NE equation then reads

$$\kappa^{\text{dis}} = \frac{e^2}{k_B T} [(2c_{\text{PS}}\omega_{\text{Li}^+} + c_{\text{TFSI}^-})D_{\text{Li}^+} + 2c_{\text{PS}}\omega_{\text{Li}^+}D_{\text{PS}} + c_{\text{TFSI}^-}D_{\text{TFSI}^-}] \quad (11)$$

Note that all of the species in this treatment are monovalent; therefore, $z_i^2 = 1$. In Figure 6a–d, we plot the conductivities κ^{dis} from eq 11 (cross symbols), which in most cases approach much closer to the experimental and MD results than the other two theoretical approaches.

Finally, all three theoretical approaches help us to elucidate the conductivity mechanism of Li_2S_x solutions. Due to high Li–PS pairing, most of the ion carriers are “neutralized” and do not contribute to the conductivity. Only the associated fraction acts on the external electric field, which results into an electric current. When LiTFSI salt is added, it contributes mostly dissociated Li^+ and TFSI^- ions and therefore fully contributes to the conductivity, which also explains the almost constant trend in Figure 6d.

As reported by Zheng et al.,⁶⁹ a practical sulfur/electrolyte (S/E) ratio (i.e., density of sulfur in electrolyte) with improved cycling stability and Coulombic efficiency is achieved for an S/E ratio of 50 g L^{-1} . This corresponds to approximately 0.4 M Li_2S_4 in our systems for a complete conversion of all sulfur into Li_2S_4 . Above this concentration, the saturation of the conductivity caused by increasing ionic pairing and viscosity can be one of the limiting factors²⁰ for using the high S/E ratio solutions.

Suo et al.⁷⁰ reported that in solvent-in-salt systems, high viscosity and incomplete solvation shell facilitate higher Li^+ transference number ($t_{\text{Li}^+} = J_{\text{Li}^+}/J$), which is unlike in conventional salt-in-solvent electrolytes. In our system, the Li^+ transference number does not increase with Li_2S_x concentration (see Figure 6e). Instead, it stays around 0.2 for a wide range of PS concentrations without LiTFSI (as also demonstrated experimentally⁶⁷). The low value can be explained by the fact that Li^+ ions are mostly moving collectively together with PS ions. The constant value of the transference number is also in accordance with the weak dependence of coordination numbers on PS concentrations. Expectedly, with 0.5 M LiTFSI, t_{Li^+} increases, since additional Li^+ that come from LiTFSI are not bound to PS and contribute to the conductivity to a greater extent. In the latter case, the contribution of PS to conductivity becomes negligible, as evident from the low PS transference number in the presence of LiTFSI (see Figure S6a,b). Again, due to the strong binding between Li^+ and S_x^{2-} , Li_2S_4 behaves as a neutral species and is therefore not subject to the electric field. We presume that these short PS chains are more likely to participate in the shuttle mechanism during the charge.

3.4. Diffusion Coefficients. Evaluating the long-time mean square displacement, we now compute the self-diffusion coefficients of all of the species in our system. The diffusion coefficients from MD simulations are compared with measured values from PFG-NMR²⁸ in Figure 8a, which suggests fair agreement. At this point, we remark that without the ECC

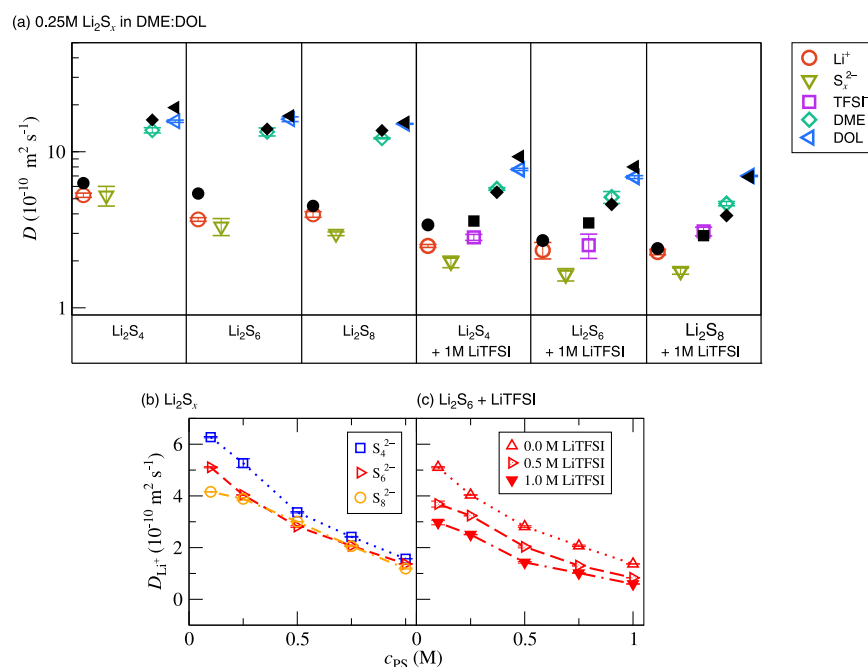


Figure 8. (a) Long-time self-diffusion coefficients D of all species in 0.25 M of the polysulfide as obtained from MD simulations (empty symbols) and PFG-NMR measurements (black filled symbols).²⁸ (b) Diffusion coefficients of Li^+ in different PS solutions as a function of PS concentration from MD simulations. (c) Diffusion coefficients of Li^+ as a function of Li_2S_6 concentration in the presence of different amounts of LiTFSI from MD simulations.

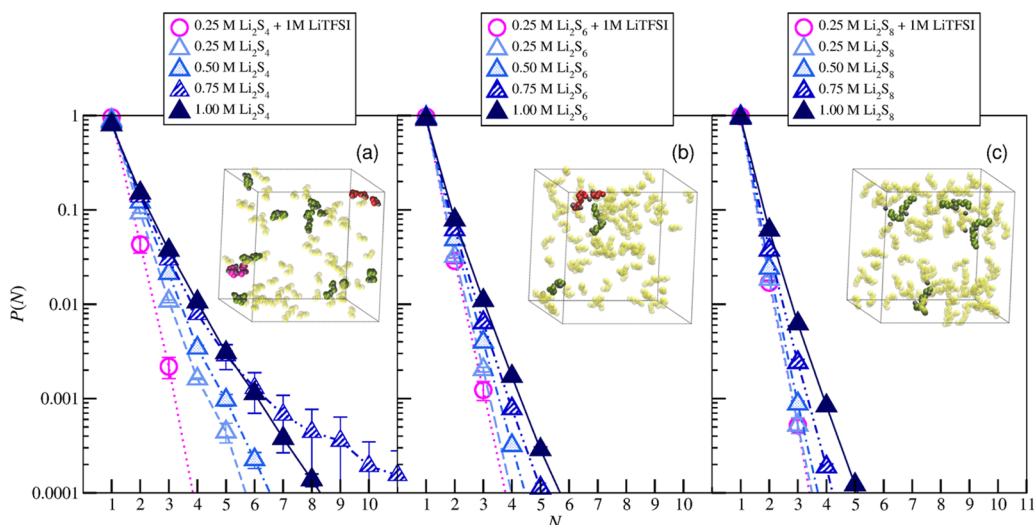


Figure 9. Cluster size distributions for (a) Li_2S_4 , (b) Li_2S_6 , and (c) Li_2S_8 in DME/DOL. The insets show representative simulation snapshots of 0.25 M of the polysulfide. Clusters of different sizes are shown in different colors: yellow ($N = 1$), green ($N = 2$), pink ($N = 3$), and red ($N = 4$).

treatment of ionic charges, the MD results would deviate from the experiments by an order of magnitude, as we demonstrate in SI (Figure S7). Namely, without ECC, the PS– Li^+ binding is unrealistically strong and thus exaggerates the ion-pair formation.^{71–74}

Shorter PS chains (S_4^{2-}) diffuse faster than longer ones (such as S_6^{2-} and S_8^{2-}), which can simply be explained in terms of the Stokes–Einstein relations, where the diffusion coefficient is inversely proportional to the particle's effective size. We furthermore note (Figure 8a) that the diffusion coefficients of Li^+ and S_x^{2-} in the solvent (without TFSI $^-$ ions) are of the same order. This can be related to collective diffusion of S_x^{2-} and Li^+ ions due to their strong association. As we increase the PS concentration, the Li^+ diffusivity monotonically

decreases (Figure 8b), as also demonstrated experimentally.⁶⁷ This effect can be ascribed to an increasing viscosity in more concentrated PS solutions (Figure S8a in SI). At low concentrations of PS, the diffusion coefficient of Li^+ significantly depends on the PS type. Namely, shorter PS (Li_2S_4) promotes higher diffusion than longer PS (Li_2S_6 and Li_2S_8). Yet, this difference disappears at higher concentrations of PS (above around 0.5 M). Interestingly, adding 1 M LiTFSI reduces the diffusion up to about 50% (Figure 8a,c), as also reported in a previous simulation study.²⁸ Also this effect can be attributed to an increased viscosity when LiTFSI is added (see Figure S8b in SI). The estimated viscosity of Li_2S_6 in DME/DOL with 1 M LiTFSI is about a factor of 3 higher than that without LiTFSI. As we have seen, viscosity plays a critical

role in ionic transport in the PS solutions. Introducing LiTFSI into dilute PS solutions increases the number of dissociated Li^+ and leads to a higher conductivity (Figure 6). Yet, this ionic effect fades out compared to an increasing viscosity at high PS concentrations.

3.5. Clustering. Due to the high attraction between terminal sulfur ends of PS and Li^+ ions, occasionally two different PS ions can bind to the same Li^+ ion such that the Li^+ ion represents a “bridging” element for the two PS ions. This can result in supramolecular clusters that are composed of several PS chains and Li^+ ions. Figure 9 shows the cluster size distribution $P(N)$ of PS in log–lin presentation. As seen, the clusters do not have a characteristic size but are extremely polydisperse and roughly follow an exponential distribution $P(N) \sim \exp(-N/\bar{N})$, where $\bar{N}$ is the mean size of the clusters.⁷⁵ Some deviations from the exponential behavior occur for larger clusters at higher concentrations, suggesting a cooperative binding. Yet, the cluster size distribution indicates that single monomers and small PS clusters prevail.

Clearly, higher concentrations of PS increase the proportion of larger clusters, simply because the probability of different PS chains to meet is higher in a more concentrated solution. Even higher concentrations, approaching the solubility limit (not shown), thus provide critical nucleation sites for the formation of large cluster precipitates. The morphology of the cluster may thus depend on the growth and nucleation rates.⁶⁹

Moreover, comparing the clustering of different PS lengths reveals that the shorter PSs tend to form larger clusters more readily, especially at high concentrations, compared to longer PSs. A higher frequency of clusters up to 80 atoms has also been observed recently for the shorter chains (S_4^{2-}) compared to the longer ones.³⁵ The reason lies in the stronger charge localization at S termini in shorter PS chains, as already discussed above. These results are in line with Vijayakumar et al.²⁴ who reported that Li_2S_4 favors dimer formation, whereas Li_2S_8 favors monomer formation in dimethyl sulfoxide solvent, while Li_2S_6 being somewhere in between. Representative simulation snapshots are shown in the insets of Figure 9, featuring larger cluster formation in the case of Li_2S_4 than in the other two cases. Note that even shorter PS chains like S_2^{2-} are even less soluble in the existing solvent and tend to precipitate out of solution even at very low concentrations^{23,61,76} (see Figure S9 in the SI for the analysis of S_2^{2-} and Figure S10 for snapshot).

Interestingly, adding 1 M LiTFSI inhibits the clustering for Li_2S_4 (Figure 9a–c), as also reported before.^{28,29} This can, however, not be claimed for the other two PS species. As discussed above, the presence of LiTFSI does not significantly influence the Li^+ –PS binding as seen from coordination numbers of S_{ter} . However, it apparently tends to inhibit cluster formation via other mechanisms, such as increased electrostatic screening due to LiTFSI ions. This influence of the ionic strength on the Li–PS network also impacts the shuttle effect in Li/S batteries.^{26,77} Sustaining the Li–S networks by using low ion-pairing salt can decrease the shuttle effect and increase the cycle performance of the batteries.^{78,79}

4. CONCLUSIONS

We developed an atomistic model for polysulfides (PS) in an organic functional solvent that is currently in use for the development of Li/S batteries. We focused particularly on the structural and dynamic properties of three different sizes of PS ions, S_4^{2-} , S_6^{2-} , and S_8^{2-} , in the presence of Li^+ and TFSI[−]

ions. The conductivity and diffusion coefficients of PS solutions are validated by experimental measurements. Conductivities of PS solutions first exhibit an increase with the Li_2S_x concentrations and eventually a saturation at around 0.5 M. The saturation in the conductivity can be linked to viscosity and the ionic correlations between PS and Li^+ , which also lead to occurrence of supramolecular clusters. The tendency of clustering increases with the concentration of Li_2S_x and is more pronounced for shorter PS ions (i.e., $x = 4$). Shorter chains have their electronic density more strongly localized at the terminal sulfur atoms than longer chains, thus facilitating the electrostatic attraction with Li^+ .

The addition of TFSI[−] ions leads to a larger amount of dissociated Li^+ ions and to a notable increase in the viscosity. The dissociated Li^+ contributes to the conductivity considerably; on the other hand, increased viscosity inhibits the conductivity at larger PS concentrations. The presence of LiTFSI also reduces cluster formation of shorter PS ions. Even though TFSI ions do not significantly reduce the Li^+ –PS binding, they weaken the binding between multiple PS ions into clusters, partially because of a higher ionic strength. Thus, LiTFSI increases the solubility of PS ions, thereby enhancing the shuttle effect in the Li/S batteries.

Our simulation results of PS solutions reveal that the structural and transport properties are subject to subtle interactions among ions and solvent molecules, which should be considered carefully when it comes to the design of electrolytes for Li/S batteries.⁸⁰ In the next step, studies on different types of anions in Li/PS solutions⁸¹ as well as in confining electrode materials are envisioned.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.8b10175.

Force-field development for sulfur nonbonded parameters; conductivity; Lennard-Jones parameterization based on conductivity; diffusion coefficients: finite-size correction (Li_2S_4 in DME/DOL); cluster analysis; radial distribution function (RDF): 0.25 M Li_2S_x in DME/DOL; integral of the correlation function; transference number; diffusion coefficients without ECC; viscosity; cluster analysis; and snapshot for Li_2S_2 in DME/DOL (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: joachim.dzubiella@physik.uni-freiburg.de.

ORCID

Chanbum Park: 0000-0002-8613-9278

Matthias Ballauff: 0000-0002-4764-2073

Matej Kanduć: 0000-0002-5307-7488

Joachim Dzubiella: 0000-0001-6751-1487

Notes

The authors declare no competing financial interest.

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